Machine Learning in Materials Chemistry

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Statement of Originality

Please note that the work presented is my own. Where AI has been used, it has been for the purpose of debugging code only.

INTRODUCTION

Problems 1 and 4 in this assignment are directly connected to lectures 2 and 3, which cover supervised learning, Gaussian Process Regression (GPR) and the use of structural descriptors in materials chemistry.

Problem 1 focuses on implementing GPR for a 1D function, following the formalism discussed in the lectures, to model the relationship between data locations ($\mathbf{x}_n$) and labels (y_n) using a kernel function $k(\mathbf{x}, \mathbf{x}')$ and kernel matrix K (see lecture 2, slide 17).

Problem 4 applies the concept of atomic structure descriptors, specifically the SOAP descriptor, to generate a 2D structure map using kernel PCA as demonstrated in lecture 2 (slide 34) and lecture 3 (slide 19).

QUESTION 1A

Please note that all the code and accompanying plots for this question can be found in the *TCC_ML>QUESTION_1* folder.

For this question, a simple function $f(x) = sin(2\pi x)$ was chosen. A description of the 1D Gaussian Process Regression workflow is given below.

1. Reference Data
   - Training inputs are given by $\{x_n\}_{n=1}^{N}$
   - Training labels are given by $\{y_n\}_{n=1}^{N}$

   In the code, these correspond to x_n and y_n respectively. They also correspond to the "Reference Data" box in Figure 4a in the *Chemical Reviews* article [1].

2. Kernel Construction
   - Kernel function $k(x, x'; \theta)$, here $k(x, x') = \sigma_f^2 exp\left[\frac{-(x-x')^2}{2l^2}\right]$
   - Hyperparameters $\theta = \{l, \sigma_f\}$

   In the code, this corresponds to the 'Build the kernel' section. This also corresponds to the "Kernel Construction" box in figure 4a.

3. Model Training
   - Kernel matrix $\mathbf{K_n n} = [k(x_i, x_j)]_{i,j=1}^{N}$
   - Regulariser $\alpha = \sigma_{noise}^2$
   - Solve for weights $\mathbf{c} = (\mathbf{K} + \alpha\mathbf{I})^{-1}\mathbf{y}$

   In the code and Figure 4a, this corresponds to the "Model Training" section.

4. Prediction
   - Test inputs $\{x_*^m\}_{m=1}^{M}$
   - Cross-kernel $\mathbf{K_{*n}} = [k(x_*^m, x_n)]_{m,n}$
   - Posterior mean $\mathbf{y}_* = \mathbf{K_{*n}c}$
   - Posterior variance $Var[y_*] = \mathbf{K_{**}} - \mathbf{K_{*n}}(\mathbf{K} + \alpha\mathbf{I})^{-1}\mathbf{K_{n*}}$

   In the code and Figure 4a, this corresponds to the "Prediction" section.

[1] Deringer et al., 2021

Kernel Choice

The choice was made to choose the squared-exponential (RBF) kernel

$$k_{\mathrm{RBF}}(x, x') = \sigma_f^2 \exp[-\tfrac{(x-x')^2}{2l^2}]$$

for its analytic simplicity, which biases the model towards very smooth functions. However, $f(x) = \sin(2\pi x)$ is perfectly periodic, so a *periodic* kernel,

$$k_{\mathrm{per}}(x, x') = \sigma_f^2 \exp\Big[-\tfrac{2\sin^2(\pi|x-x'|\,/\,p)}{l^2}\Big],$$

or a composite kernel such as RBF $\times$, periodic would capture the oscillatory structure by construction and require fewer datapoints to fit the full amplitude.

Discussion of Results

Running the code for Question 1(a) generates the plots in Figures 1 and 2.

Figure 1: Plot of the training set data

In Figure 1, the training data is randomly generated within the interval $x \in [0, 3]$ at a noise level of $\sigma_{noise} = 0.1$. As expected, the training data falls neatly within the true function $f(x) = sin(2\pi x)$.

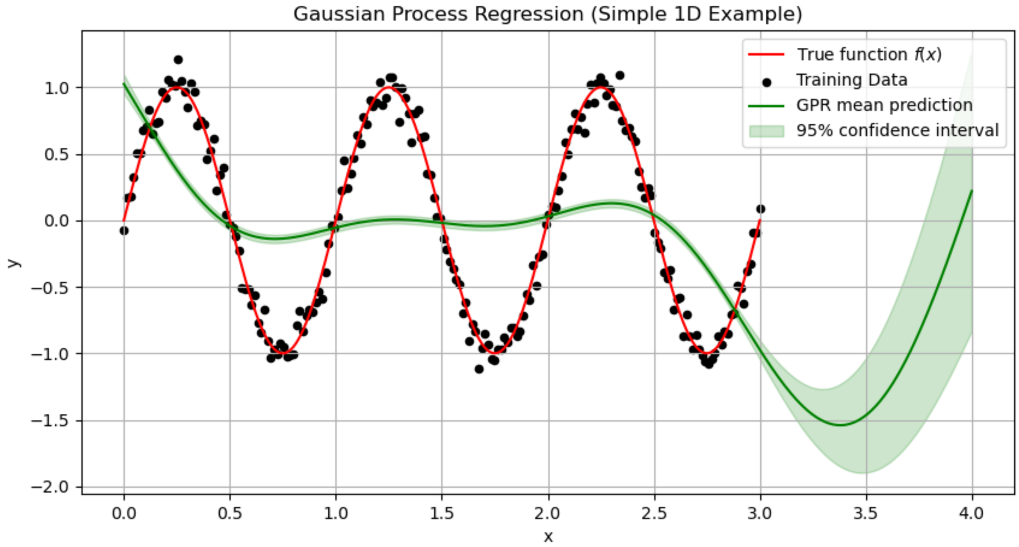

Figure 2: Plot of the predicted data

Within the training region $x \in [0, 3]$ in Figure 2, the GPR is most confident where data are densest. As the prior mean is zero and length-scale $l = 1$, the posterior mean remains near zero and fails to capture the sine oscillations of period 1. At the edges ($x \approx 0.3$), asymmetry in the data causes the mean to deviate toward $sin(2\pi x)$. Beyond $x = 3$, the rapidly widening band highlights the model's uncertainty in extrapolation.

Limitations and Further Considerations

While the RBF-GP scales to higher dimensions in concept, its implementation costs $\mathcal{O}(N^3)$ and becomes unmanageable for large datasets. In practical applications, one must therefore

- Optimise hyperparameters,
- Use sparse approximations to reduce the cubic cost to $\mathcal{O}(NM^2)$ or better,
- Carefully select (or engineer) kernels to encode known physics or periodicities

Moving forward, one could also draw posterior samples to visualise uncertainty, split off a test set to compute RMSE and then extend the same workflow to high-dimensional SOAP descriptors in Problem 4.

QUESTION 4A

Please note that all the code and accompanying plots for this question can be found in the *TCC_ML>QUESTION_4* folder.

For this question, the data in a paper titled *"Automated generation of structure datasets for machine learning potentials and alloys"* was used [2]. A description of the workflow is given below.

1. Data Loading
   The data ASE database file was opened and all structures were loaded.
2. Species identification
   All structures were scanned to find out which elements appear. These elements were then sorted and passed into DScribe's SOAP constructor.
3. SOAP 'fingerprint'
   Each structure is represented by the average of its atom-centred SOAP power spectra. The hyper-parameters were chosen as follows:
   - Cutoff radius, $r_{cut} = 5.0$ Å, captures both the first and second coordination shells in typical small-molecule and condensed-phase chemistries while avoiding inclusion of far-field atoms that contribute noise and increase computational cost.
   - Radial basis size, $n_{max} = 8$, provides enough resolution to distinguish variations in bond lengths without exploding the vector length.
   - Angular momentum, $l_{max} = 6$, encodes detailed directional information (e.g. tetrahedral vs. trigonal coordination).
   - Dense output (sparse = False) was used because the resulting vector is efficiently handled in NumPy and scikit-learn.
4. Random sampling
   The total dataset contains more than 116.000 structures. Trying to visualise the structures using a 2D structure map results in 'bus' errors and as such, a random sample of N = 15,000 structures has been used for this problem.
5. Kernel PCA
   A linear KernelPCA is fitted to the sampled vectors and the top two components are retained.
6. Visualisation
   Each structure's projection (PC1, PC2) was plotted as a point.

Descriptor and Kernel-PCA Formalism

Lectures 2 & 3 (Slides 34 & 19) were followed by representing each structure via its atom-centred SOAP power spectrum averaged over atoms:

$$\bar{\mathbf{p}} = \frac{1}{N_{\text{atoms}}} \sum_{i=1}^{N_{\text{atoms}}} \{p_{nn'l}(i)\}$$

where $p_{nn'l}(i)$ is the spherical harmonic + radial basis power spectrum for centre i. Hyperparameters $r_{cut} = 5$Å, $n_{max} = 8$ and $l_{max} = 6$ were chosen to balance the resolution of first and second coordination shells against descriptor dimensionality.
These averaged vectors $\bar{\mathbf{p}_k}$ are then reduced to 2D via linear Kernel-PCA:

$$k_{ij} = \bar{\mathbf{p}_i} \cdot \bar{\mathbf{p}_j},$$

with centre K, and solved for $K, \mathbf{v} = \lambda, \mathbf{v}$, retaining the two eigenvectors of largest λ as PC1 and PC2.

[2] Poul et al., 2025

Discussion of the Structure Map

Figure 3 reveals a number of things:

1. A *dense central cluster*, containing the bulk of small-molecule and crystalline motifs with similar local coordination environments.

2. Three *protruding wings*, each corresponding to families of structures like elemental phases or high-coordination clusters.

3. A few *potential outliers* such as the point around $(-180, 300)$, arising from rare or highly strained geometries.

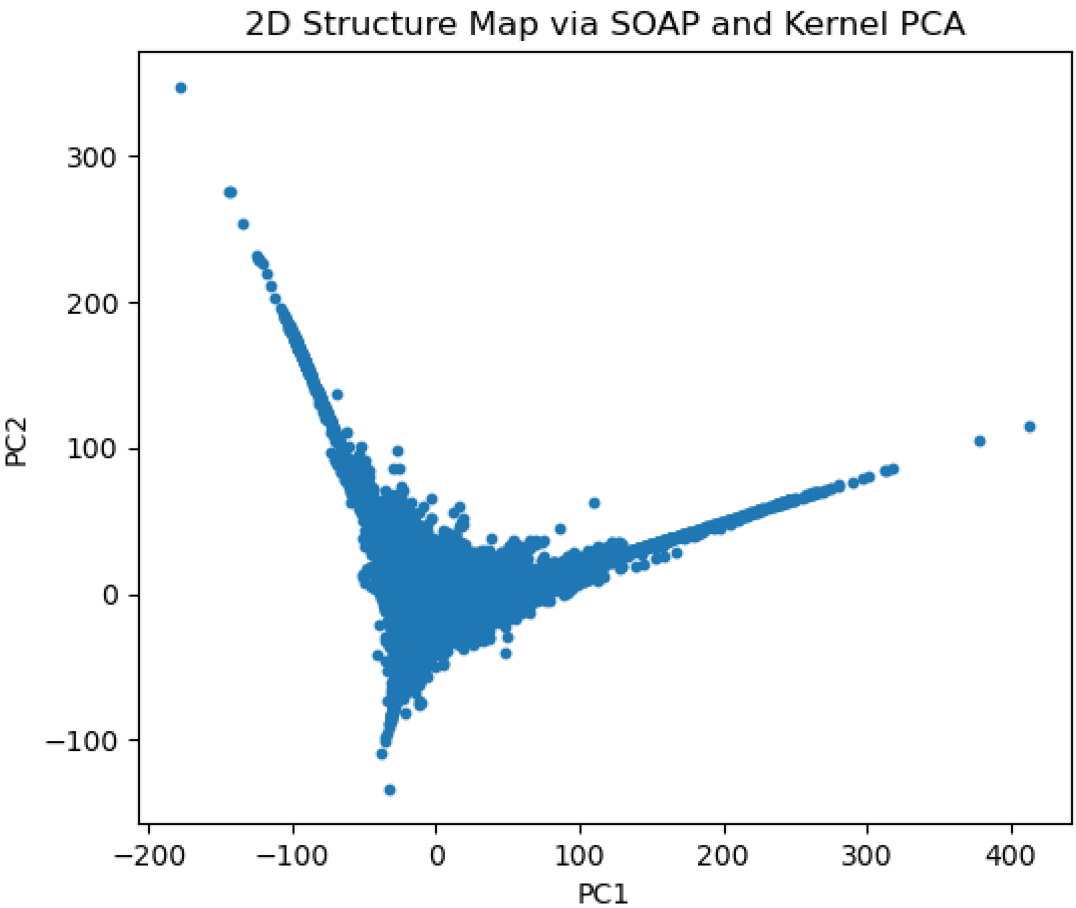

Figure 3: 2D Structure Map

Limitations and Further Considerations

- **Sampling bias**: Randomly selecting 15.000 of 116.000 structures may under-represent rare but chemically important environments. Sampling by composition or energy would give a more balanced map.

- **Linear kernel**: While fast, linear Kernel-PCA can only separate those environments lying on a linear subspace. A non-linear (e.g. RBF) kernel PCA or t-SNE might uncover finer clusters.

- **Computational cost**: Computing full SOAP for large datasets scales poorly. Sparse-SOAP approximations could be explored in the future to bypass this problem.

References

1. Deringer, V. L., Bartók, A. P., Bernstein, N., Wilkins, D. M., Ceriotti, M., & Csányi, G. (2021). Gaussian Process Regression for Materials and Molecules. In Chemical Reviews (Vol. 121, Issue 16, pp. 10073–10141). American Chemical Society. https://doi.org/10.1021/acs.chemrev.1c00022

2. Poul, M., Huber, L., & Neugebauer, J. (2025). Automated generation of structure datasets for machine learning potentials and alloys. Npj Computational Materials, 11(1). https://doi.org/10.1038/s41524-025-01669-4